UPLOAD SLOT 05 OF 12

3D & Spatial Visualizations

Presentation boards and illustrative views. Renders are artistic impressions; the analysis outputs are computed.

CONTENTS

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository	**github.com/wasim437/AI_SAFA**
Live portal	**wasim437.github.io/AI_SAFA/**
Produced by	**src/boards.py** · **src/plan.py** · **archive/withdrawn_visuals/README.md**
Reproduce	`python run_analysis.py` · `python -m tests.test_pipeline`

Al Safa 2 Park — Falaj Al Safa

Mohamed Wasim · Individual Applicant

Generated 2026-08-03 · reproducible: python tools/build_submission_pdfs.py

How this document was produced

This submission is the output of a ten-phase process in which each phase is checked against the one before it. Nothing downstream is hand-drawn or hand-typed: change an input, re-run, and every chart, drawing and figure moves with it — or a test fails loudly. The phases behind this particular document are marked.

Phase 1 — Site & Context Analysis
: 39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

Phase 2 — Problem Definition
: Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

Phase 3 — Opportunity & Objectives
: The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

Phase 4 — Concept Development
: Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

Phase 5 — Masterplan Development
: Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

Phase 6 — Detailed Design
: The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

Phase 7 — Performance & Sustainability
: Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

Phase 8 — User Experience & Activation
: K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

Phase 9 — AI Workflow & Visualisation
: The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

Phase 10 — Submission Assembly
: Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

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THE CODE AND DATA BEHIND THIS DOCUMENT

src/boards.py
src/plan.py
archive/withdrawn_visuals/README.md

REPRODUCE IT

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 38 checks
```

3D & Spatial Visualisations

What each image is, what produced it, and which ones are evidence rather than illustration

Three of this project's images were withdrawn for presenting invented data as measurement, and six photoreal renders were withdrawn for showing a park that is not the one in the drawings. What follows from that is a rule applied to every visual here: no image appears without stating what produced it and what it may be used to conclude.

Upload slot 05 — 3D & Spatial Visualisations
Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
Mohamed Wasim · Individual Applicant
Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. Three classes of image, never mixed

Class	What it is	What it may be used for
Technical drawing	Generated from the plan geometry in `src/plan.py` and drawn to scale	Measuring. Dimensions and areas taken off it are correct because they are the drawn polygon's own dimensions
Analysis output	Computed from project data by the pipeline — a chart of a model's behaviour or a simulation's result	Reading a result. It reports what the model found, at the accuracy stated on the figure
Artistic impression	AI-generated illustration of the design language	Judging character and atmosphere only. It is not evidence of anything, and no number may be taken from it

Every image in this submission carries its class on its own sheet. A juror should never have to guess which they are holding.

2. What is in this slot

Image	Class	Produced by
Board 1 — Concept	Presentation board	src/boards.py, reading src/plan.py
Board 2 — Evidence	Presentation board	src/boards.py, reading the trained-model metrics
Masterplan	Technical drawing	src/plan.py — areas are the shoelace area of each polygon

3. The photoreal renders, and why this slot does not currently contain any

Six photoreal renders were produced earlier in the project. Each was opened and compared against the masterplan, and all six failed: one showed a serpentine canopy over a large lagoon, one a dead-straight corridor — the superseded scheme the crescent replaced — one a vaulted pavilion planted with tropical species that contradict the five-species desert palette, and the remainder curved as an S rather than as one arc. All six were withdrawn.

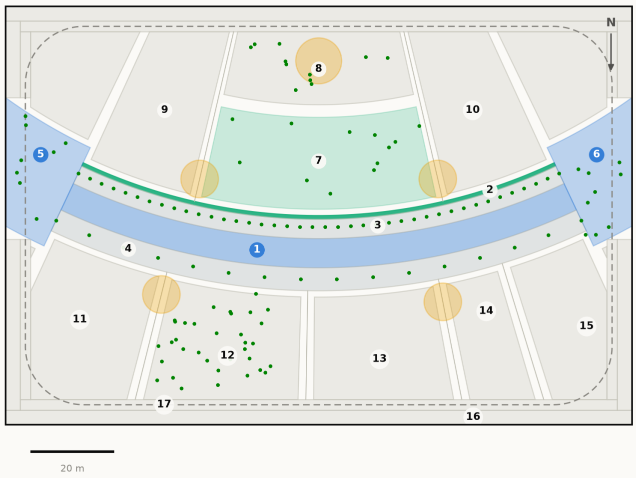
The design they were supposed to show is specific and measurable: one continuous arc of 141 m radius bowing convex south, a 0.9 m water channel at its northern drip line, and 18 m of canopy over a 7 m walk. A render that shows a lake where the design has a rill is not a stylistic difference. It contradicts the sustainability argument, in which the entire water surface of the park is 105 m² — 0.70% of the site.

This slot is therefore presented with its measured drawings and its two boards, and without photoreal imagery, until renders exist that pass the acceptance test published with this project. A submission that shows a juror two different parks forfeits the authority of everything else in it. Board 1 marks the two views still in preparation rather than filling them with an image that would have to be captioned dishonestly.

4. The test every image must pass before it is admitted

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
	Al Sikkak — shaded alleys & setbacks	1,414 m²
	Total	**15,000 m²**

Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

The masterplan every visualisation in this submission must agree with. Technical drawing — to scale.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

Board 1 concept

Presentation board.

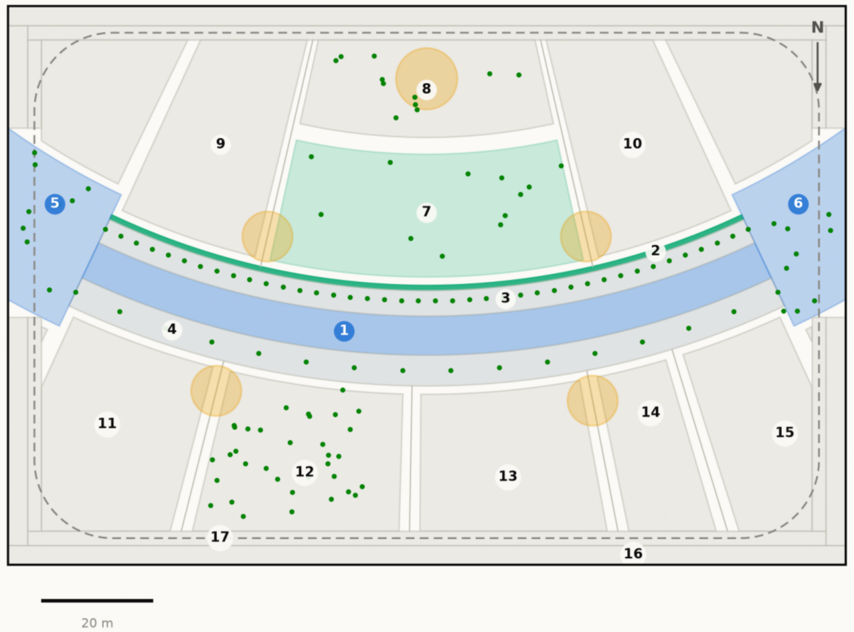
AL SAFA 2 PARK — FALAJ AL SAFA

Dubai Municipality AI Park Design Challenge | Board 1 of 2 — CONCEPT

The concept, the plan, and the section that makes it work

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



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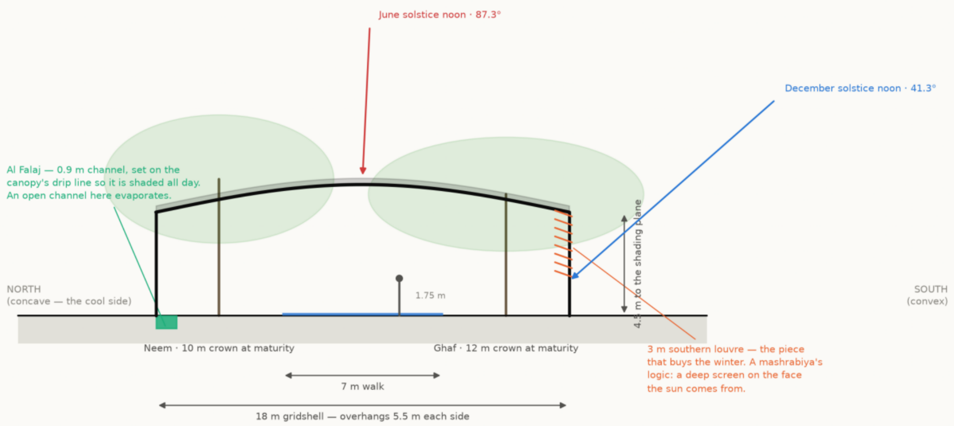
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Section A–A — the Crescent Canopy at midspan

Every dimension read from config.CRESCENT; sun angles computed by the NREL algorithm at 25.190°N



The plane alone loses the walk to a low southern sun. The louvre is what holds the shadow on the path from November to January.
Source: src/config.py CRESCENT; sun angles NREL SPA via pvlib · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

The crescent from the north-west, at golden hour

Visualisation in preparation

Aerial view · AL_SAFA_MASTER_PROMPT.md prompt 01

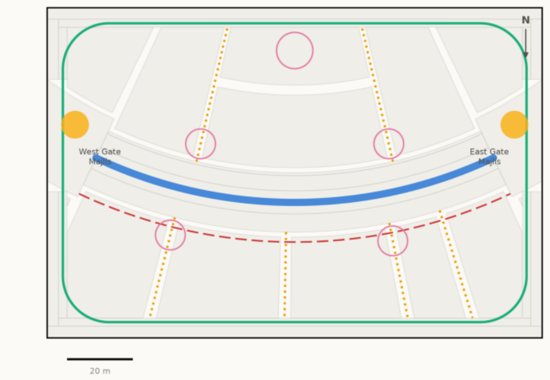
Beneath Al Hilal — the perforated soffit at eye level

Visualisation in preparation

Eye-level view · AL_SAFA_MASTER_PROMPT.md prompt 03

Circulation & accessibility

One shaded primary route, radial alleys off it, and a running loop around the whole



Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped.
Source: src/plan.py — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

One arc. Every room in the park is struck off its centre, so no room is a rectangle and every room faces the crescent square-on.
The bow is 18 m deep on a 141 m radius — swept against the 8,760-hour solar model, not drawn by eye. A straight canopy shades marginally more ground on average and leaves the walk with no shade at all for 330 hours a year; this one leaves it for 56.
Walk shaded 87.3% of daylight hours · 131 trees · 15,000 m² · every number reproducible with `python run_analysis.py`

Board 2 evidence

Presentation board.

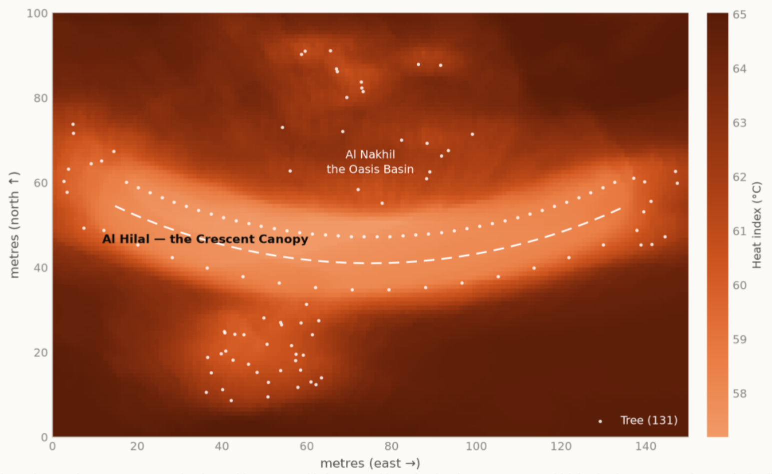
AL SAFA 2 PARK — FALAJ AL SAFA

Dubai Municipality AI Park Design Challenge | Board 2 of 2 — EVIDENCE

What the models measured, and what they changed about the design

Predicted summer comfort across the site

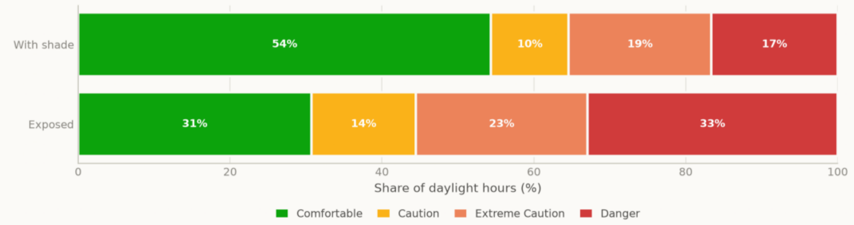
July afternoon heat index per square metre — the shade strategy, tested



Lighter is cooler. The crescent reads as the coolest continuous route across the site, and its concave side is measurably cooler than its convex one — which is why the rooms people are asked to linger in are all placed there. White dots are the 131 trees.
Source: NREL Solar Position Algorithm via pvlib, 25.190N 55.238E; NCM Dubai climate normals 1977-2015 (39-year record) · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Shade converts unusable daylight hours into usable ones

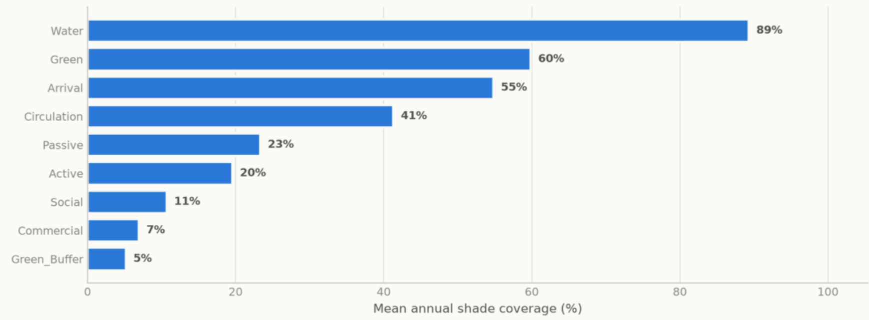
Share of the year's 4,402 daylight hours in each comfort band



Source: NCM Dubai climate normals 1977-2015 (39-year record); comfort bands DERIVED via NWS heat index · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Annual shade coverage by zone type

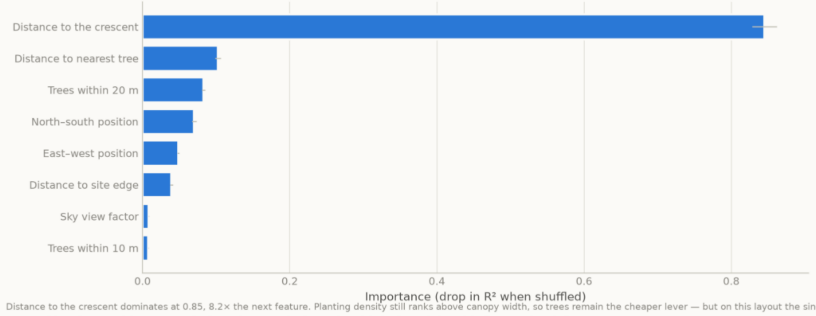
Ray-traced ground-plane occlusion, 1 m grid, sampled across the year



Source: NREL Solar Position Algorithm via pvlib, 25.190N 55.238E; canopy and tree geometry from the Phase 5/6 design · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

What actually determines whether a square metre is shaded

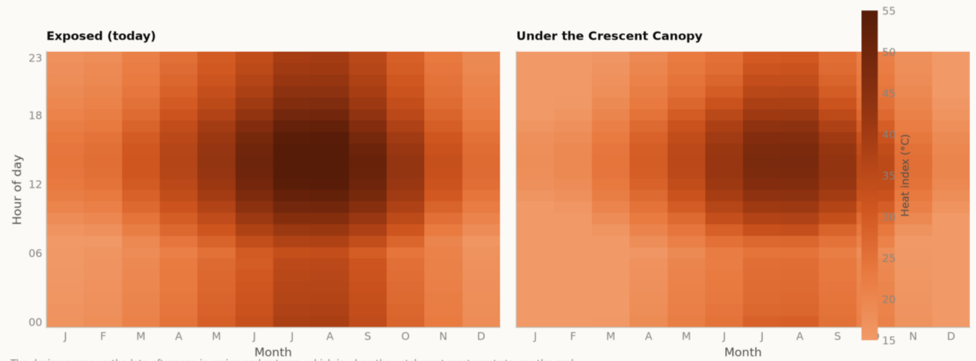
Permutation importance — the drop in accuracy when each feature is shuffled



Distance to the crescent dominates at 0.85, 8.2× the next feature. Planting density still ranks above canopy width, so trees remain the cheaper lever — but on this layout the single strongest predictor of whether a square metre is shaded is simply how far it is from the crescent.
Source: This project's models — see notebooks/ and src/ · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

When is the park comfortable?

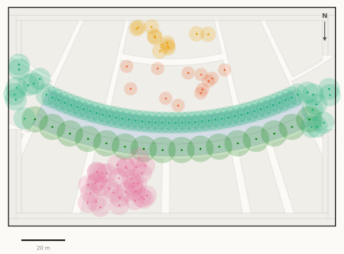
Mean heat index by hour and month — exposed today, shaded as designed



The design opens up the late afternoon in spring and autumn, which is when the catchment most wants to use the park.
Source: NCM Dubai climate normals 1977-2015 (39-year record) — hourly series MODELLED, see src/climate.py · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Planting plan — 131 trees

Drawn at mature canopy radius. The avenue is structural: it carries the shoulder seasons the gridshell cannot

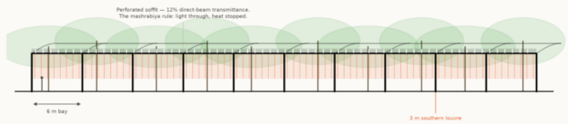


- Ghaf (Prosopis cineraria) ×16 · 45 L/day
- Neem (Azadirachta indica) ×58 · 90 L/day
- Ficus nitida (Ficus microcarpa nitida) ×34 · 110 L/day
- Olive (Olea europaea) ×11 · 60 L/day
- Date Palm (Phoenix dactylifera) ×12 · 120 L/day

28 of 131 trees are UAE natives (Ghaf and Date Palm). Peak summer irrigation demand 11.8 m³/day. Ghaf takes the southern rank because it is the most drought-tolerant species in the schedule and that rank is the more exposed of the two.
Source: data/raw/species_water_carbon_rates.csv; layout from src/solar.py · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Elevation — the Crescent Canopy

60 m of the 124 m run, to scale. 6 m structural bay

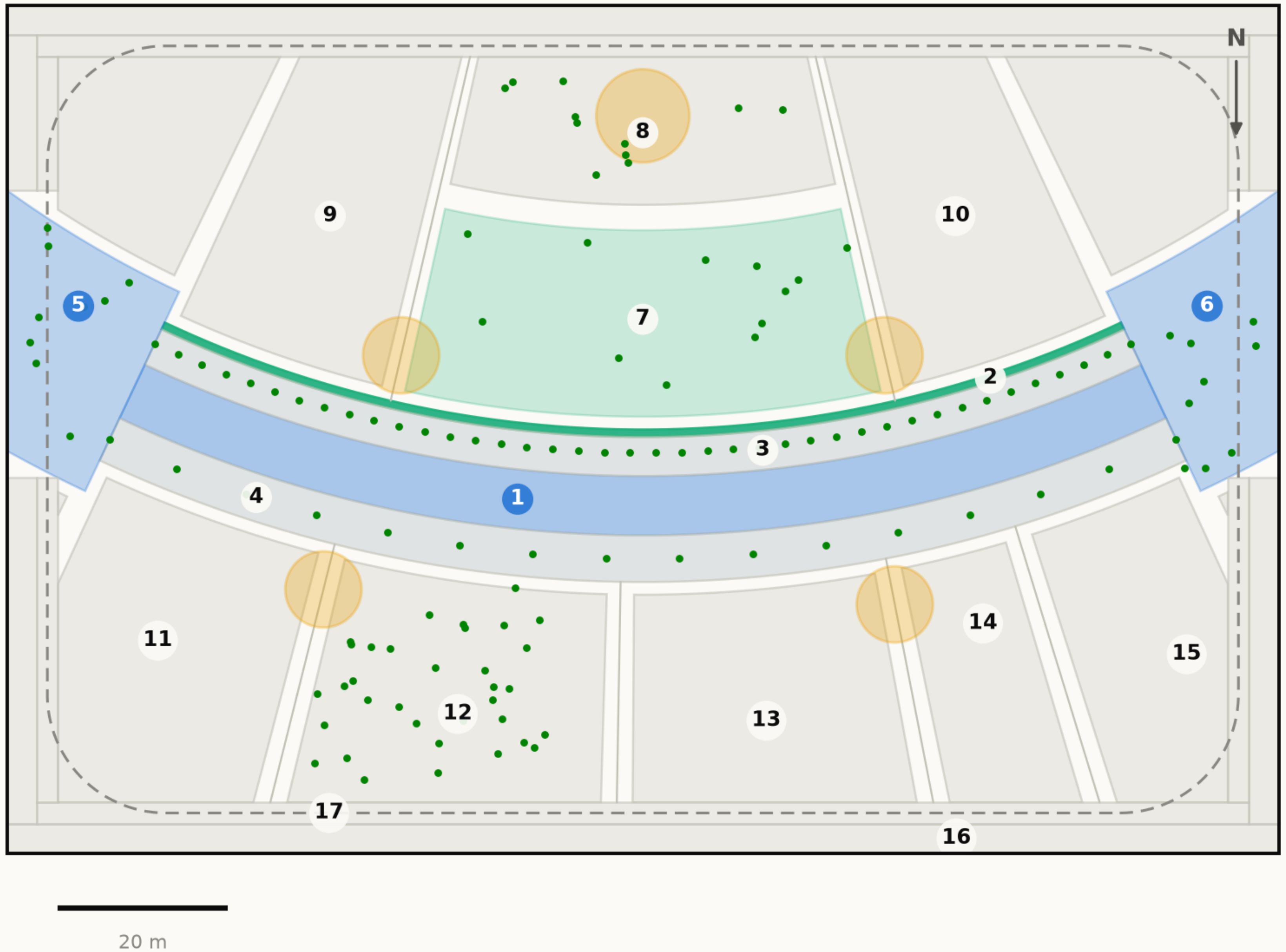


21 bays over the full run. Because the plan is an arc, no two bays are identical in plan — but every one is the same section, which is what keeps it buildable. Trees at mature canopy from the Phase 6 planting schedule.
Source: src/config.py CRESCENT; bay spacing from the Phase 6 structure · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Comfortable daylight hours rise from 44.5% to 64.6% — 20.1 points of the year handed back. Peak heat index falls 56.8 °C → 48.7 °C.
Site-wide mean shade is 34.1%, and that is stated as a position rather than hidden: this scheme makes a few places genuinely excellent instead of the whole site marginally better.
Shade surrogate R^2 0.994 · comfort classifier 97.5% with temperature and humidity withheld · 2 microclimate regimes, k selected by silhouette

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



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Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

The complete project, and where to check it

Every link below is live. The repository holds the data, the code, the trained models and the tests, and runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository	**github.com/wasim437/Al_SAFA**
Live portal	**wasim437.github.io/Al_SAFA/**

START HERE

EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language
PROJECT_PLAN.md	Requirements, phases, status, what is left
README.md	The design argument, written for a juror
notebooks/AL_SAFA_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded

THE GEOMETRY AND THE MODELS

src/plan.py	Single source of the crescent geometry
src/climate.py	The 8,760-hour year rebuilt from NCM normals
src/solar.py	Sun position and shadow ray-tracing
src/dataset.py	Assembles the ML training tables
src/models.py	The four models
src/costing.py	The cost model against the AED 35 M ceiling

THE DATA, AS ISSUED AND AS PROCESSED

data/raw/sources.json	Every source dataset, with its period
DATA_SOURCES.md	Sources and their stated limitations
data/raw/site_zoning_schedule.csv	The measured room schedule
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/cost_plan.csv	The capital cost plan, line by line

THE RESULTS, AND THE CHECKS ON THEM

models/model_metrics.json	Trained-model metrics
models/headline_metrics.json	The headline numbers
tests/test_pipeline.py	38 correctness checks
archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why

THE DRAWINGS

figures/fig10_masterplan.png	Masterplan and room schedule
design/visuals/section_crescent.png	Section A–A at midspan
design/visuals/circulation_crescent.png	Circulation and accessibility
design/visuals/facilities_crescent.png	Commercial & Service Facilities Map
design/visuals/planting_crescent.png	Planting plan, 131 trees

Links resolve once the repository is published. GitHub renders PDFs and notebooks in the browser — nothing needs to be downloaded or cloned.